

Matteo Pettenò



Profile

AI Researcher and Software Engineer specializing in deep learning, audio processing and controllable generative models. Formerly a Marie Skłodowska-Curie Doctoral Fellow within the Privacy for Smart Speech Technology network (jointly hosted by EURECOM and Ruhr-Universität Bochum), I published research on voice anonymization alongside my prior academic work in controllable music generation. I hold an M.Sc. in Music & Acoustic Engineering (Politecnico di Milano), a B.Sc. in Information Engineering (University of Padova), and multi-year industry experience as a Full-Stack Developer, DevOps Engineer, and Software Architect. I am fully available for full-time roles in AI/ML and software engineering.

Education

Ph.D. Candidate in PSST (Marie Curie Fellow)

09/2025 - 02/2026

Digital Security Department, EURECOM

Sophia Antipolis, France

IKA, Ruhr-Universität Bochum (RUB)

Bochum, Germany

- *Privacy for Smart Speech Technology* (PSST) joint doctoral training program
- Funded by Horizon Europe Marie Skłodowska-Curie Action
- 19 months at EURECOM, 6-months secondment at Orange and 26 months at RUB
- **Research focus:** Disentangled representations for selective attribute suppression

Master's Degree in Music and Acoustic Engineering

09/2021 - 10/2024

DEIB, Politecnico di Milano (PoliMi)

Milan, Italy

- **Relevant Courses:** Machine Learning, Computer Music, Sound Analysis Synthesis and Processing, Creative Programming and Computing, Musical Acoustics, Electronics and Electroacoustics, Computer Security
- **Thesis:** *Latent Space Regularization via Normalizing Attribute Transformations for Symbolic Music Generation*
- **Advisor:** Alberto Bernardini
- **Grade:** Cum Laude (110/110)

Bachelor's Degree in Information Engineering

09/2013 - 07/2021

DEI, University of Padua (UNIPD)

Padua, Italy

- **Relevant Courses:** Algorithms for Engineering, Systems and Models, Control systems, Electronics, Telecommunications
- **Thesis:** *Evaluation of the performance of commercial STT and NER services applied to digitized oral sources*
- **Advisor:** Sergio Canazza Targon
- **Grade:** 93/110
- **Note:** Suspension of studies from 2016 to 2019.

High School Certificate in Information Technology

09/2007 - 07/2013

I.T.I.S. Carlo Zuccante

Mestre, Italy

- **Relevant Courses:** Computer Science, Electronics, Systems, Statistics
- **Thesis:** *The MIDI Protocol & Touchscreen XY MIDI Controller*
- **Grade:** 87/100

Publications

S. Chandra, **M. Pettenò**, N. Evans et al. "Evaluating voice anonymisation using similarity rank disclosure", in Proc. of the *Odyssey - The Speaker and Language Recognition Workshop*, Lisbon, Portugal, June, 23-26, 2026

M. Pettenò, A. I. Mezza and A. Bernardini, "Conditional Diffusion As Latent Constraints for Controllable Symbolic Music Generation", in Proc. of the *26th Conference of the International Society for Music Information Retrieval (ISMIR)*, Daejeon, Korea, Sept. 21-25, 2025

M. Pettenò, A. I. Mezza and A. Bernardini, "On the Joint Minimization of Regularization Loss Functions in Deep Variational Bayesian Methods for Attribute-Controlled Symbolic Music Generation", in Proc. of the *33rd European Signal Processing Conference (EUSIPCO)*, Palermo, Italy, Sept. 8-12, 2025.

R. B. Luzietti et. al, "FONTI 4.0: Evaluating Speech-to-Text Automatic Transcription of Digitized Historical Oral Sources". *Proceedings of the Eighth Italian Conference on Computational Linguistics CliC-it*, 2021¹

¹Although my name is mentioned only in the acknowledgments and not among the authors, this publication consists of much of the work done for my bachelor's degree thesis.

Professional Experience**Full Stack Developer**

11/2021 - 08/2023

ccelera s.r.l (Arsenalia Group) - Via Lepetit, 8, 20124

Milan, Italy

- **Description:** full-stack development on the SAP Hybris Commerce platform.
- **Platforms:** SAP Hybris Commerce
- **Customers:** Bonfiglioli, Cellularline, PegPerego, Metal Work
- **Job type:** Part-time

DevOps Engineer

09/2019 - 02/2021

Walit s.r.l - Via Dandolo, 25/B, 31100

Treviso, Italy

- **Description:** Design and maintenance of the Google Cloud infrastructure for a deep learning project. CI/CD orchestration in Gitlab. Application security monitoring with OWASP ZAP.
- **Platforms:** Google Cloud Platform (GCP)
- **Job type:** Full-time

System Integration Engineer

01/2019 - 07/2019

Alpenite Ltd - 38 Craven Street, WC2N 5NG

London, UK

- **Description:** architect/developer role in collaboration with an Indian team for analysis and development of logistics and retail integration on Mulesoft middleware.
Integration strategies used: AMQP messaging protocol (RabbitMQ as message-broker), REST/SOAP API calls to ERP M3, reading/writing files on FTP.
- **Platforms:** Mulesoft
- **Customers:** Stella McCartney
- **Job type:** Full-time

Full Stack Developer

01/2017 - 01/2019

Alpenite s.r.l (Arsenalia Group) - Via delle Industrie, 27/7, 30175

Venice, Italy

- **Description:** full-stack development for B2B and CRM on the SAP Hybris Commerce platform. CRM: integration of the IFM Group phone bar, ASM (Assisted Service Module), ticketing and notification system, asynchronous integration of the RFC SAP call for prices and availability of goods on the ZK framework (event queue system). B2B: multi-login function for users linked to multiple customers, Authorization Matrix framework (functionality/site content visibility BE control), mobile channel development on PWA with Workbox library and site content management setup with CMS.
- **Platforms:** SAP Hybris Commerce
- **Customers:** Kering Eyewear

	• Job type: Full-time	
	IT Help Desk Operator <i>Alfa Telematica s.r.l. - Via Antonio Pacinotti, 4, 30175</i>	09/2014 - 01/2017 Venice, Italy
	• Customers: Regione Veneto • Job type: Part-time	
Personal Projects	Unbaised <i>Co-Creator/Lead Developer</i>	03/2025 - 11/2025 Remote, Venice (Italy) / San Francisco (USA)
	• Description: Co-designed and engineered an AI-driven media platform allowing users to control news generation through bias metrics (e.g. political spectrum alignment). The system autonomously ingests daily multi-source news, synthesizes parameter-conditioned summary articles and dynamically generates custom audio podcasts. • Tech Stack: Python, Docker (Microservices), Google Cloud Platform (GCP), Expo (Mobile Frontend), Tavily API, RSS Feeds, ElevenLabs. • Job type: Independent Startup Concept / Remote Collaboration	
	Homelab <i>Self-Hosted Homelab & Smart Home</i>	2022 - Present
	• Description: Architected and actively maintain a Proxmox VE hypervisor running self-hosted services across LXC containers and VMs. Deployed media ecosystems (*arr stack, Jellyfin, Audiobookshelf), network storage (OMV, Nextcloud, PhotoPrism), and networking/security (Caddy reverse proxy, Pi-hole, WireGuard VPN, Dante SOCKS proxy, Gitea). Engineered an automated smart irrigation system managed via Home Assistant over a custom Zigbee mesh network.	
	Around the Job <i>Full Stack Developer & Creator</i>	2023 - Present github
	• Description: Built an interactive web application to map and navigate target company locations during my job search. Features an interactive map interface connected to a cloud database to explore companies based on preferred living locations. • Tech Stack: Vue.js, Leaflet, Firebase, Firestore.	
	Advent of Code <i>Challenge Solutions & API</i>	2022 - Present link
	• Description: Solves annual December algorithmic challenges with strict constraints (no external libraries/help). Implemented a Flask-based web server deployed on Render to accept daily puzzle input files and automatically compute and return solutions. • Tech Stack: Python, Flask, Render, REST API, Algorithms & Data Structures.	
Research Interests	Deep Learning, Representation Learning, Speech Processing, Privacy Preservation, Speaker Anonymization, Deepfake Detection, Music Information Retrieval (MIR), Audio Generation, AI-Assisted Music Composition, Music Understanding, Hearing Aids	
Research Projects	Evaluating voice anonymisation using similarity rank disclosure <i>In Proc. of the Odyssey 2026 - The Speaker and Language Recognition Workshop</i> Keywords: voice anonymisation, privacy, evaluation Traditional voice anonymisation evaluation relies heavily on EER, which provides an incomplete view of privacy risk. We introduce Similarity Rank Disclosure (SRD), a threshold-independent, information-theoretic metric that assesses privacy directly at the representation level. Applied to speaker embeddings, fundamental frequency, and phone representations from 2024 VoicePrivacy Challenge systems, SRD uncovers worst-case privacy leaks and system vulnerabilities missed by standard EER metrics.	2026

Conditional Diffusion as Latent Constraints for Unconditional Symbolic Music Generation Models 2025

In Proc. of the 26th Conference of the ISMIR

[github](#)

Keywords: symbolic music, attribute-controlled generation, diffusion models, latent constraints

A plug-and-play framework using conditional denoising diffusion models as latent constraints for symbolic music generation. The approach enables fine-grained control over musical attributes such as note density, pitch range, contour, and rhythm complexity, without retraining the backbone model. Demonstrated superior controllability and quality compared to traditional attribute regularization methods.

On the Joint Minimization of Regularization Loss Functions in Deep Variational Bayesian Methods for Attribute-Controlled Symbolic Music Generation 2025

In Proc. of the 33rd EUSIPCO

[github](#)

Keywords: symbolic music, attribute-controlled generation, power transforms

Supervised extension of the well-established variational information bottleneck framework by including a regularization loss aimed at encoding a specific attribute into a designated dimension of the latent space. The novelty of the approach lies in the introduction of an invertible parametric mapping whose goal is to transform the original and possibly complex distribution of the target attribute to a new one that is coherent with that of the latent space. The dataset generated to train the models is publicly available on Zenodo.

Do Unconditional Deep Generative Models Spontaneously Learn How to Encode Human-Interpretable Musical Attributes? 2023

Music and Acoustics Engineering Capstone course in MS.

[github](#)

Keywords: variational autoencoders, latent space topological structure

Latent space analysis of the pre-trained recurrent variational autoencoder model MusicVAE by Google Magenta in its 2-bars melody unconditional version. The study is focused on investigating the presence of any correlations between the topological structure of the latent space and different human-interpretable musical attributes of the output.

Evaluation of the performance of commercial STT and NER services applied to digitized oral sources 2021

Thesis in Information Engineering BS

[github](#)

Keywords: speech-to-text, named-entity-recognition, gcp, aws

This thesis is part of the Fonti 4.0 project and it aims to test the performances of STT commercial services on digitized historical oral sources in order to discover which one is the most accurate and find out if the quality of the digitized audio or other elements in the recording impact the accuracy of automatic transcription. In the experiment the transcription services were evaluated through some metrics obtained aligning and comparing the reference manual transcription with the automatic transcription and a web interface has been implemented that allows you to explore how they change these indices vary according to the quality of the audio, the language spoken and the presence or absence of disturbing events. Furthermore, a first step is taken to study the NER algorithms.

Creative Projects

Ego 2023

Creative Programming & Computing course in MS

[github](#)

Keywords: three.js, glsl, svelte, mediapipe, max4live, tone.js

Creative coding project meant to explore the idea of human perception, in particular the idea of identity and self consciousness and the way it is distorted and biased.

Pulseq - Fractal Sequencer 2022

Advanced Coding Tools and Methodologies course in MS

[github](#)

Keywords: fractal sequencer, web app, svelte, tone.js, glsl

Fractal sequencer implemented in Svelte as a single-page application (SPA) and inspired by Eurorack module Bloom by Qu-Bit Electronix.

Computer Music Projects

Padder - Computer Music System 2022

Computer Music Languages and Systems course in MS

[github](#)

Keywords: arduino, touchosc, supercollider, processing

Musical instrument based on a virtual pedal that, thanks to the interaction between the different blocks of its architecture, can generate chords that can be played in real time during a musical performance. This kind of approach could be seen as a handsfree way to generate chords, instead of using a piano or a keyboard.

OranJam - JUCE 2022

Computer Music Languages and Systems course in MS

[github](#)

Keywords: juce, c++, cmake

Polyphonic subtractive synthesizer implemented in JUCE. Features: oscillators bank, white noise generator, ADSR envelope, filters bank and LFO.

HarMMMLonizer - Supercollider 2022

Computer Music Languages and Systems course in MS

[github](#)

Keywords: supercollider, harmonizer, delay lines, crosstalk delay feedback

Real-time harmonizer implemented in Supercollider. It features a controllable number of voices and a delay line for each voice with three different feedback configurations, including a crosstalk one for stereo ping-ping effects.

Template Based Chord Recognition 2021

Computer Music Representations and Models course in MS

[github](#)

Keywords: MIR, chord recognition, librosa, libfmp

Propose and implement a metric able to evaluate the performance of the template based chord recognition algorithm. The proposed metric is then used to evaluate the performance on 4 different songs. Finally, the evaluation is repeated for different values of the algorithm parameters.

Rhythmic and Harmonic Analysis 2021

Computer Music Representations and Models course in MS

[report](#)

Keywords: music theory

Rhythmic analysis of the song *Pyramid Song* by Radiohead and harmonic analysis of the jazz standard *Peace* by Horace Silver in the arrangement of Norah Jones.

Sound Analysis Synthesis and Processing Projects

Wave Digital Filter Modeling 2022

Sound Synthesis and Spatial Processing course in MS

[report](#)

Keywords: wdf, matlab, virtual analog

Design of a three-way crossover network in the Wave Digital (WD) domain starting from a reference analog circuit. The model is then implemented in MATLAB using the trapezoidal discretization method (bilinear transformation).

Leslie Speaker Emulation 2022

Sound Synthesis and Spatial Processing course in MS

[report](#)

Keywords: leslie speaker, matlab, digital audio effect

Efficient implementation of the Leslie rotary speaker as a digital audio effect.

Acoustic Source Localization with Microphone Array 2022

Digital Audio Analysis and Processing course in MS

[report](#)

Keywords: sound localization, doa estimation, matlab, microphone arrays

Acoustic source localization using the Delay-And-Sum (DAS) beamformer and the MUSIC methods to estimate the Direction Of Arrival (DOA) of two audio sources sampled by a 64 microphones array.

RIR Estimation with Wiener Filters 2022

Digital Audio Analysis and Processing course in MS

report

Keywords: room impulse response, wiener filter, matlab, convolution

Estimation of the Room Impulse Response (RIR) of a small reverberant environment by means of a Wiener filter. The obtained filter is obtained using the Overlap-and-Add (OLA) algorithm.

Musical Acoustics Projects

Design of a Recorder Flute 2023

Musical Acoustics: Characterization of Musical Instruments course in MS

report

Keywords: applied acoustics, matlab, flute modeling

Design of a recorder flute in MATLAB dimensioning the bore, the last two finger holes, the flue channel and the instrument mouth.

Brass Instrument Simulation 2023

Musical Acoustics: Characterization of Musical Instruments course in MS

report

Keywords: applied acoustics, comsol, trumpet modeling

Implementation of a trumpet model in COMSOL in order to simulate its acoustic response.

Design of a Piano 2023

Musical Acoustics: Characterization of Musical Instruments course in MS

report

Keywords: applied acoustics, comsol, matlab, piano modeling

Design and analysis of a piano soundboard and its bridge in COMSOL.

Horn Design 2022

Musical Acoustics: Modeling of Musical Instruments course in MS

report

Keywords: applied acoustics, matlab, horns modeling

Design of an exponential and compound horns in MATLAB.

Musical Instruments Modeling Techniques 2022

Musical Acoustics: Modeling of Musical Instruments course in MS

report

Keywords: applied acoustics, matlab, piano modeling, guitar modeling

Implementation in MATLAB of the FD model for a piano string considering the hammer interaction and of an acoustic guitar model with 20 resonances.

Helmholtz Resonator Tree 2022

Musical Acoustics: Modeling of Musical Instruments course in MS

report

Keywords: applied acoustics, helmholtz resonator, matlab, simulink

Modeling of a complex resonant system through a hierarchical structure of Helmholtz resonators.

Glass Harp 2022

Musical Acoustics: Modeling of Musical Instruments course in MS

report

Keywords: applied acoustics, comsol, glass harp modelling

3D and axysymmetric modeling of a wineglass for glass harp in COMSOL with eigenfrequencies analysis.

2D and 1D systems 2022

Musical Acoustics: Modeling of Musical Instruments course in MS

report

Keywords: applied acoustics, matlab

Circular membrane and circular plate characterization in MATLAB. Analysis of the interaction in a system formed by a circular plate coupled with an iron string.

Helmholtz Resonator and System Impedance 2022

Musical Acoustics: Modeling of Musical Instruments course in MS

report

Keywords: applied acoustics, helmholtz resonator, matlab

Analysis of the characteristics of a Helmholtz resonator in MATLAB.

Additional Experience	Volunteer at 24th ISMIR Conference <i>Politecnico di Milano</i> • Assisted with event coordination and attendee registration • Supported speakers and managed technical equipment • Helped ensure smooth operations during sessions and networking events	November, 2023 Milan, Italy
Digital Skills	Programming Languages: Python, C++, MATLAB, GLSL, JAVA, JS, CSS ML/DL: Keras, Tensorflow, PyTorch, Google Magenta, NumPy, SciPy, scikit-learn Big Data Tools: Apache Beam, Apache Airflow, Apache Spark Scientific Tools: LaTeX, Jupyter Notebooks, Zotero MIR: librosa, FMP Notebooks DevOps: GCP, AWS, CI/CD, Docker, LXC, Proxmox Web: Tone.js, Three.js, Svelte, Vue.js, Firebase, Flask, Spring, PWA, Workbox, Hugo Computer Music Languages: JUCE, Supercollider, Faust, PureData DAWs: Logic Pro, Ableton Live, Reaper, Ardour Acoustics: COMSOL, REW Cybersecurity: gdb, OWASP ZAP, ghidra DBMS: Microsoft SQL Server, MySQL, SQLite, MongoDB, Redis, MariaDB Design: Figma Operating Systems: Linux, macOS, Windows, Hackintosh, Android Others: Blockchain	
Hobbies	Music: Guitarist and mixing/recording engineer for my band @venus_in.cancer. Outdoors: Nature enthusiast. Experienced in solo and group wild bikepacking journeys and mountain trekking. DIY Projects: Creative builder with a resource-driven mindset. Built a custom wood shed, handcrafted a balance board, upcycled an antique family bicycle into a coaster-brake bike, experimented with homebrewing and many more. Sports: Lifelong basketball player. Gardening: Dedicated gardener maintaining my home garden.	
Languages	Italian: Mother tongue English: Fluent (C1) French: Base (A1)	
References	Available upon request.	